

Laboratorio 5

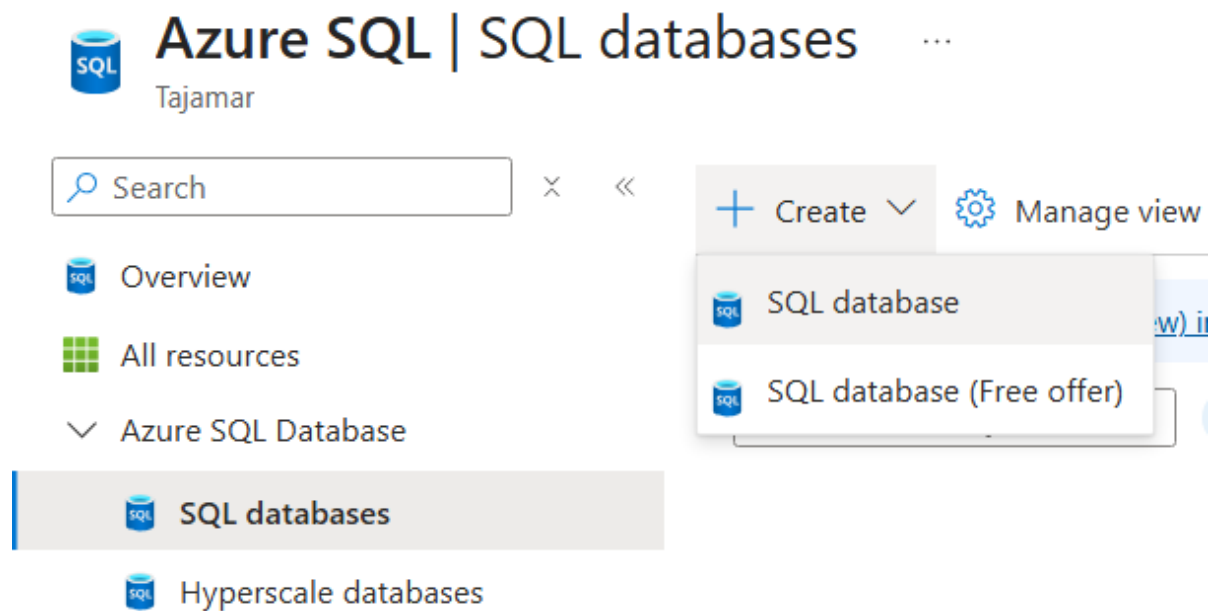


Lab 5

Provision an Azure SQL Database

Create an Azure SQL Database

Home > Azure SQL



Azure SQL | SQL databases ...

Tajamar

Search

Overview

All resources

▼ Azure SQL Database

SQL databases

Hyperscale databases

+ Create ▼ Manage view

- SQL database
- SQL database (Free offer)

Create SQL Database Server

Home > Azure SQL | SQL databases > Create SQL Database

Create SQL Database Server ...

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *	server-lab5-alumno09 ✓
	.database.windows.net
Location *	(Europe) Spain Central ▼

Authentication

 Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#) 

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication [Learn more](#)  using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) 

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication [Learn more](#)  using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) 

Authentication method	☐ Use Microsoft Entra-only authentication
	☐ Use both SQL and Microsoft Entra authentication
	☒ Use SQL authentication
Server admin login *	admin-lab5-alumno09 ✓
Password *	 ✓
Confirm password *	 ✓

Create SQL Database

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

 Create a serverless Azure SQL Database with 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. Limit is 10 free databases per subscription. [Learn more](#)

Apply offer

 SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure for Students

Resource group * ⓘ

(New) resourcegroup-lab5-alumno09

[Create new](#)



Cost summary

General Purpose (GP_S_Gen5_1)

Cost per GB (in USD) 0.14

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.69 USD

COMPUTE COST / VCORE SECOND ¹ 0.000159 USD

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

[Home](#) > [Azure SQL | SQL databases](#)

Create SQL Database ...

Microsoft

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

SecurityLabDB ✓

Server * ⓘ

(new) server-lab5-alumno09 (Spain Central) ▼

[Create new](#)

Want to use SQL elastic pool? ⓘ

☐ Yes ☒ No

Workload environment

☒ Development
☐ Production

i Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ⓘ

General Purpose - Serverless

Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled

[Configure database](#)

Backup storage redundancy

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

The default backup storage redundancy setting is taken from the setting of the source.

Backup storage redundancy ⓘ

☒ Locally-redundant backup storage
☐ Zone-redundant backup storage[Review + create](#)[Next : Networking >](#)

Home > Azure SQL | SQL databases

Create SQL Database ...

Microsoft

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'server-lab5-alumno09' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method * ⓘ

- ☐ No access
- ☒ Public endpoint
- ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

Home > Azure SQL | SQL databases

Create SQL Database ...

Microsoft

- Basics
- Networking
- Security
- Additional settings
- Tags
- Review + create

Product details

SQL database
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost
Storage cost 5.69 USD / month + Compute cost 0.000159 USD / vCore second

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Azure for Students
Resource group	resourcegroup-lab5-alumno09
Region	Spain Central
Database name	SecurityLabDB
Server	(new) server-lab5-alumno09

Create

< Previous

[Download a template for automation](#)

Create sample tables

Plugins

The screenshot displays the Visual Studio Code interface with the Extensions Marketplace open. On the left, a list of extensions is shown, including 'SQL Bindings', 'SQL Server (mssql)', 'SQLTools SQL Server', 'Azure Data Lake Tools', 'SQL Server Profiler Tool', 'SQL Server Client (mssql)', 'SQL Wayfarer', 'SQL File Detection Tool', and 'maxdop - T-SQL formatter'. The 'SQL Server (mssql)' extension is selected and its details are shown on the right.

SQL Server (mssql)
Microsoft | microsoft.com | 9,791,102 | ⭐⭐⭐☆☆ (206)
Design and optimize schemas for SQL Server, Azure SQL, and SQL Database in Fabric using a modern, lightweight extension built for developers.

Details: FEATURES, CHANGELOG, DEPENDENCIES, EXTENSION PACK

⚙️ Roadmap 2026 | 🐛 Report Bug | 🗨️ Request Feature | 💬 discussions: 58 total | 🧪 Build and Test (Unit + E2E) | 🟢 passing | 📊 codecov: 82%

MSSQL extension for Visual Studio Code

The **MSSQL Extension for Visual Studio Code** is designed to empower developers by providing a seamless and modern database development experience. Our goal is to make SQL development more productive and intuitive by integrating essential features such as schema management, query execution, and AI-powered assistance.

Explore and Learn

Microsoft

Get started with the MSSQL extension for VS Code

Marketplace

Identifier	ms-mssql.mssql
Version	1.46.8
Published	10 years ago
Last Released	1 hour ago

Categories

Programming Languages | Azure

Resources

- Repository
- Issues
- License
- Microsoft
- Marketplace

Connect Database

 **Connect to Database**

Profile Name

SecurityLabProfile

Connection Group

<Default> 

Input type

☒  Parameters [Load from Connection String](#)

☐  Browse Azure

☐  Browse Fabric

Server name * 

server-lab5-alumno09.database.windows.net

Port

1433

Trust server certificate 

☐

Authentication type * 

SQL Login 

User name * 

admin-lab5-alumno09

Password * 

..... 

Save Password

☒

Database name 

SecurityLabDB 

Encrypt 

Mandatory 

Advanced 

Test connection

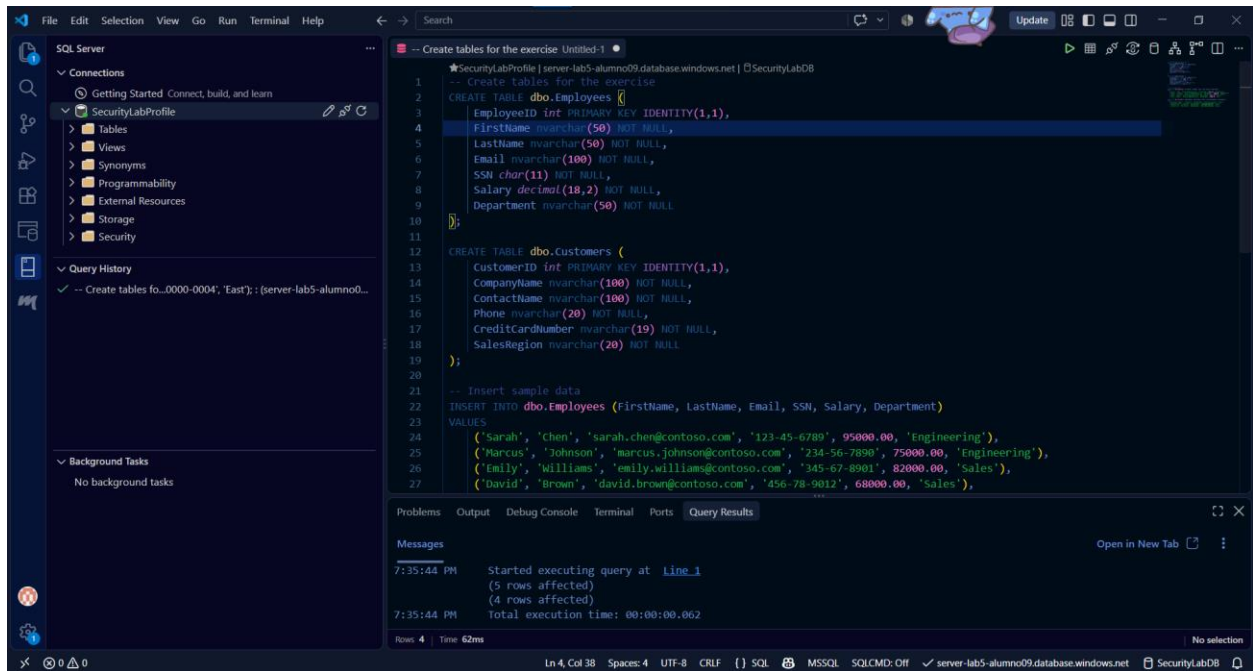
Connect 

 Import

Saved Conne

Connect with the current settings and save the connection profile

Create Tables



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'SQL Server' tree is expanded to 'SecurityLabProfile', showing 'Tables', 'Views', 'Synonyms', 'Programmability', 'External Resources', 'Storage', and 'Security'. The 'Query History' pane shows a query titled '-- Create tables to...0000-0004', 'East': (server-lab5-alumno0...'. The 'Background Tasks' pane shows 'No background tasks'. The main query window, titled '-- Create tables for the exercise: Untitled-1', contains the following SQL code:

```
-- Create tables for the exercise
1  CREATE TABLE dbo.Employees
2  (
3      EmployeeID int PRIMARY KEY IDENTITY(1,1),
4      FirstName nvarchar(50) NOT NULL,
5      LastName nvarchar(50) NOT NULL,
6      Email nvarchar(100) NOT NULL,
7      SSN char(11) NOT NULL,
8      Salary decimal(18,2) NOT NULL,
9      Department nvarchar(50) NOT NULL
10 );
11
12 CREATE TABLE dbo.Customers (
13     CustomerID int PRIMARY KEY IDENTITY(1,1),
14     CompanyName nvarchar(100) NOT NULL,
15     ContactName nvarchar(100) NOT NULL,
16     Phone nvarchar(20) NOT NULL,
17     CreditCardNumber nvarchar(19) NOT NULL,
18     SalesRegion nvarchar(20) NOT NULL
19 );
20
21 -- Insert sample data
22 INSERT INTO dbo.Employees (FirstName, LastName, Email, SSN, Salary, Department)
23 VALUES
24 (
25     'Sarah', 'Chen', 'sarah.chen@contoso.com', '123-45-6789', 95000.00, 'Engineering'),
26 (
27     'Marcus', 'Johnson', 'marcus.johnson@contoso.com', '234-56-7890', 75000.00, 'Engineering'),
28 (
29     'Emily', 'Williams', 'emily.williams@contoso.com', '345-67-8901', 82000.00, 'Sales'),
30 (
31     'David', 'Brown', 'david.brown@contoso.com', '456-78-9012', 68000.00, 'Sales'),
32 );
```

The 'Messages' pane at the bottom shows the following output:

```
7:35:44 PM Started executing query at Line 1
7:35:44 PM (5 rows affected)
7:35:44 PM (4 rows affected)
7:35:44 PM Total execution time: 00:00:00.062
```

The status bar at the bottom indicates 'Ln 4, Col 38 Spaces: 4 UTF-8 CRLF {} SQL MSSQL SQLCMD: Off ✓ server-lab5-alumno09.database.windows.net SecurityLabDB'.

Implement Dynamic Data Masking

The screenshot displays the SQL Server Enterprise Manager interface with two windows open, illustrating the implementation of Dynamic Data Masking.

Top Window: SQL Script

```

-- Create tables for the exercise
1  -- Mask SSN to show only last 4 digits
2  ALTER TABLE dbo.Employees
3  ALTER COLUMN SSN ADD MASKED WITH (FUNCTION = 'partial(0, "XXX-XX-", 4)');
4
5  -- Mask Salary with a random value
6  ALTER TABLE dbo.Employees
7  ALTER COLUMN Salary ADD MASKED WITH (FUNCTION = 'random(50000, 150000)');
8
9  -- Mask Email to show first character and domain
10 ALTER TABLE dbo.Employees
11 ALTER COLUMN Email ADD MASKED WITH (FUNCTION = 'email()');
12
13 -- Mask credit card to show only last 4 digits
14 ALTER TABLE dbo.Customers
15 ALTER COLUMN CreditCardNumber ADD MASKED WITH (FUNCTION = 'partial(0, "XXXX-XXXX-XXXX-", 4)');
16
17 -- Mask phone number to show only last 4 digits
18 ALTER TABLE dbo.Customers
19 ALTER COLUMN Phone ADD MASKED WITH (FUNCTION = 'partial(0, "XXX-XXX-", 4)');
20
21 -- Create a user without DBMS permission
22 CREATE USER MaskedViewer WITHOUT LOGIN;
23
24 -- Grant SELECT on dbo.Employees to MaskedViewer;
25 GRANT SELECT ON dbo.Employees TO MaskedViewer;
26
27 -- Grant SELECT on dbo.Customers to MaskedViewer;
28 GRANT SELECT ON dbo.Customers TO MaskedViewer;
29
30 -- Query as the masked user (data appears masked)
31 EXECUTE AS USER = 'MaskedViewer';
32 SELECT FirstName, LastName, Email, SSN, Salary FROM dbo.Employees;
33 SELECT CompanyName, ContactName, Phone, CreditCardNumber FROM dbo.Customers;
34 REVERT;
35
36 -- Query as admin (data appears unmasked)
37 EXECUTE AS USER = 'sa';
38 SELECT FirstName, LastName, Email, SSN, Salary FROM dbo.Employees;
39 REVERT;

```

Bottom Window: Query Results

Query 1: Masked Data (EXECUTE AS USER = 'MaskedViewer')

FirstName	LastName	Email	SSN	Salary
Sarah	Chen	c000@XXXX.com	XXX-XX-6789	89919.36
Marcus	Johnson	m000@XXXX.com	XXX-XX-7890	66985.85
Emily	Williams	e000@XXXX.com	XXX-XX-8901	83637.75
David	Brown	d000@XXXX.com	XXX-XX-9012	118051.83
Lisa	Garcia	l000@XXXX.com	XXX-XX-0123	63656.30

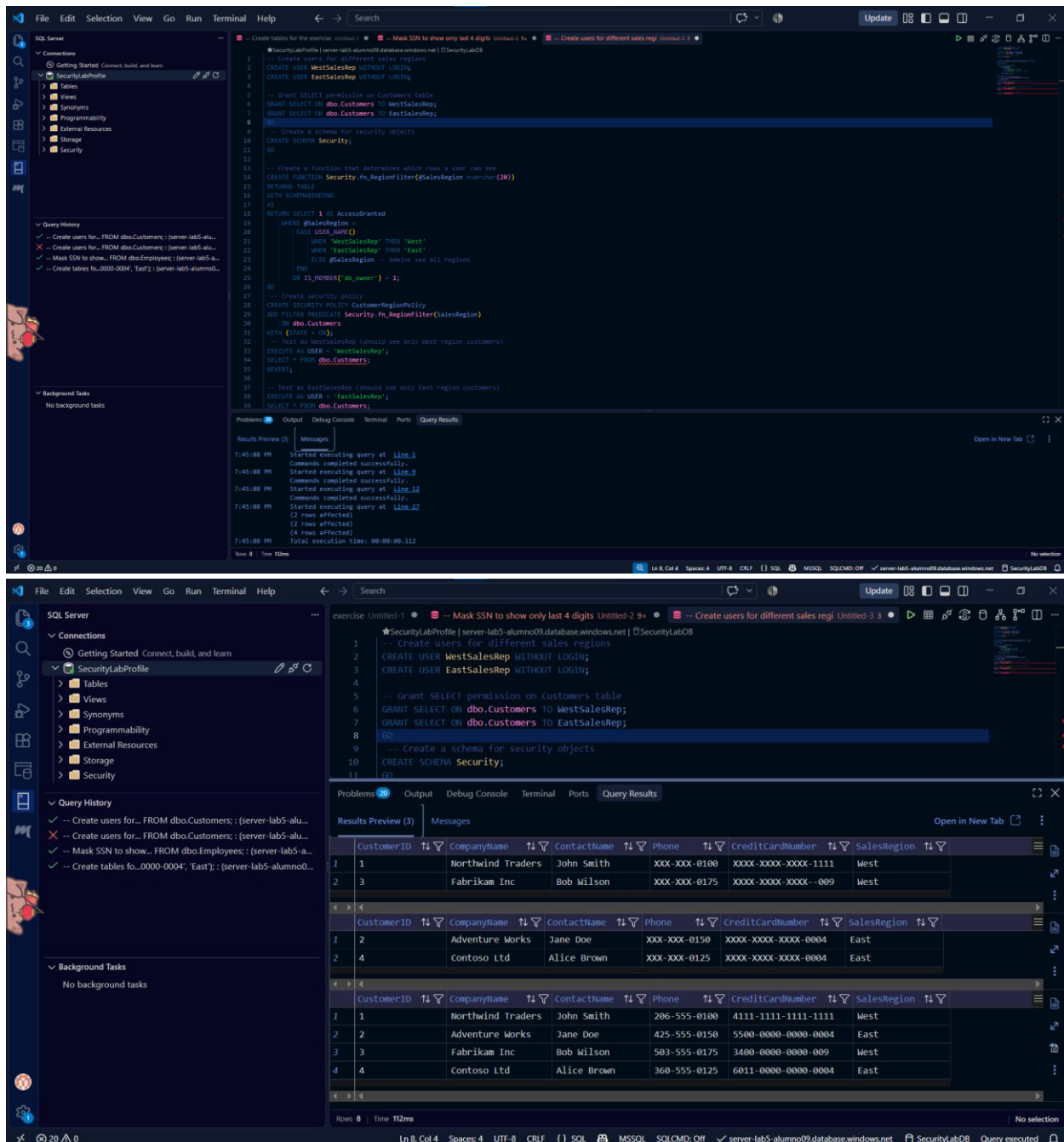
Query 2: Masked Data (EXECUTE AS USER = 'sa')

Companyname	ContactName	Phone	CreditCardNumber
Northwind Traders	John Smith	XXX-XXX-0100	XXXX-XXXX-XXXX-1111
Adventure Works	Jane Doe	XXX-XXX-0150	XXXX-XXXX-XXXX-0004
Fabrikam Inc	Bob Wilson	XXX-XXX-0175	XXXX-XXXX-XXXX-009
Contoso Ltd	Alice Brown	XXX-XXX-0125	XXXX-XXXX-XXXX-8804

Query 3: Unmasked Data (EXECUTE AS USER = 'sa')

FirstName	LastName	Email	SSN	Salary
Sarah	Chen	sarah.chen@contoso.com	123-45-6789	95000.00
Marcus	Johnson	marcus.johnson@contoso.com	234-56-7890	75000.00
Emily	Williams	emily.williams@contoso.com	345-67-8901	82000.00
David	Brown	david.brown@contoso.com	456-78-9012	68000.00
Lisa	Garcia	lisa.garcia@contoso.com	567-89-0123	71000.00

Implement Row-Level Security



The screenshot displays the SQL Server Enterprise Manager interface, showing the implementation of Row-Level Security (RLS) on the Customers table. The top panel shows the SQL script being executed, and the bottom panel shows the query results.

SQL Script (Top Panel):

```
-- Create tables for the exercise (unittest 1)
-- Mask SSN to show only last 4 digits (unittest 2)
-- Create users for different sales regions (unittest 3)

-- Create users for different sales regions
CREATE USER WestSalesRep WITHOUT LOGIN;
CREATE USER EastSalesRep WITHOUT LOGIN;

-- Grant SELECT permission on Customers table
GRANT SELECT ON dbo.Customers TO WestSalesRep;
GRANT SELECT ON dbo.Customers TO EastSalesRep;

-- Create a schema for security objects
CREATE SCHEMA Security;
GO

-- Create a function that determines which rows a user can see
CREATE FUNCTION Security.fn_RegionFilter(@SalesRegion nvarchar(20))
RETURNS TABLE
WITH SCHEMABINDING
AS
RETURN SELECT 1 AS AccessGranted
WHERE @SalesRegion <=
CASE USER_NAME()
    WHEN 'WestSalesRep' THEN 'West'
    WHEN 'EastSalesRep' THEN 'East'
    ELSE @SalesRegion -- Admins see all regions
END
OR IS_MEMBER('db_owner') = 1;
GO

-- Create security policy
CREATE SECURITY POLICY CustomerRegionPolicy
ADD FILTER PREDICATE Security.fn_RegionFilter(SalesRegion)
ON dbo.Customers
WITH (STATE = ON);
GO

-- Test as WestSalesRep (should see only West region customers)
EXECUTE AS USER = 'WestSalesRep';
SELECT * FROM dbo.Customers;
REVERT;
GO

-- Test as EastSalesRep (should see only East region customers)
EXECUTE AS USER = 'EastSalesRep';
SELECT * FROM dbo.Customers;
REVERT;
GO
```

Query Results (Bottom Panel):

The query results show the Customers table filtered by SalesRegion. The results are displayed in two sections, each showing 4 rows.

CustomerID	CompanyName	ContactName	Phone	CreditCardNumber	SalesRegion
1	Northwind Traders	John Smith	XXX-XXX-0100	XXXX-XXXX-XXXX-1111	West
2	Adventure Works	Jane Doe	XXX-XXX-0150	XXXX-XXXX-XXXX-0004	East
3	Fabrikam Inc	Bob Wilson	XXX-XXX-0175	XXXX-XXXX-XXXX-0009	West
4	Contoso Ltd	Alice Brown	XXX-XXX-0125	XXXX-XXXX-XXXX-0004	East

The bottom section of the results shows the same data, but with the SalesRegion column filtered to 'West'.

CustomerID	CompanyName	ContactName	Phone	CreditCardNumber	SalesRegion
1	Northwind Traders	John Smith	206-555-0100	4111-1111-1111-1111	West
2	Adventure Works	Jane Doe	425-555-0150	5500-0000-0000-0004	East
3	Fabrikam Inc	Bob Wilson	503-555-0175	3400-0000-0000-0009	West
4	Contoso Ltd	Alice Brown	360-555-0125	6011-0000-0000-0004	East

Cleanup

The screenshot displays two windows used for database cleanup. The top window is SQL Server Enterprise Manager, showing a query script to remove security policies, functions, users, and tables. The bottom window is the Microsoft Azure portal, showing the 'Delete a resource group' dialog for 'resourcegroup-lab5-alumno09'.

SQL Server Query Script:

```
-- Remove security policy and function
1  -- Remove security policy and function
2  DROP SECURITY POLICY IF EXISTS CustomerRegionPolicy;
3  DROP FUNCTION IF EXISTS Security.fn_RegionFilter;
4  DROP SCHEMA IF EXISTS Security;
5
6  -- Remove test users
7  DROP USER IF EXISTS MaskedViewer;
8  DROP USER IF EXISTS WestSalesRep;
9  DROP USER IF EXISTS EastSalesRep;
10
11 -- Drop tables
12 DROP TABLE IF EXISTS dbo.Customers;
13 DROP TABLE IF EXISTS dbo.Employees;
```

Azure Portal Dialog: Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted: resourcegroup-lab5-alumno09

2 resources being deleted across 1 subscriptions, 1 resource groups
All dependent resources, including hidden types, are shown

Name	Resource type	Subscription	Resource Group
server-lab5-alumno09	SQL server	Azure for Students	resourcegroup-lab5-alumno09
master (server-lab5-alumno09/m...)	SQL database	Azure for Students	resourcegroup-lab5-alumno09

Enter resource group name to confirm deletion *

resourcegroup-lab5-alumno09

Delete **Cancel**